

Module 6 challenge: System Administration Consultation

Question 1:

Learning Goals:

1. Use the systems administration concepts you learned in the course to provide technical improvements to current processes.
2. Implement solutions based on an organization's restrictions, like financial resources, number of users, etc.

Overview: You'll take what you learned in the System Administration and IT Infrastructure Services course and apply that knowledge to real-world situations.

Assignment: For this writing project, you'll be presented with three scenarios for different companies. You'll be doing the systems administration for each company's IT infrastructure. For each scenario, present improvements to processes based on the company's needs and current restrictions. There's no right or wrong answer to your consultation, but your responses should explain the problem, the improvement, and the rationale behind them. Please write a 200-400 word process review for each company presented to you.

Scenario 1:

You're doing systems administration work for Network Funtime Company. Evaluate their current IT infrastructure needs and limitations, then provide at least five process improvements and rationale behind those improvements. Write a 200-400 word process review for this consultation. Remember, there's no right or wrong answer, but make sure to provide your reasoning.

Software Company:

Network Funtime Company is a small company that builds open-source software. The company is made up software engineers, a few designers, one person in

Human Resources (HR), and a small sales team. Altogether, there are 100 employees. They recently hired you as a system administrator to come in and become their IT department.

When a new person is hired on, the HR person purchases a laptop for them to do their work. The HR representative is unfamiliar with what type of hardware is out there; if a new employee requests a laptop, the HR person will purchase the cheapest option for a laptop online. Because of this, almost everyone has a different laptop model. The company doesn't have too much revenue to spend, so they don't order laptops until someone gets hired at the company. This leads to a few days of wait time from when someone starts to when they can actually work on a laptop.

The company doesn't label their computers with anything, so if a computer is missing or stolen, there's no way to audit it. There's no inventory system to keep track of what's currently in the fleet.

Once a computer is purchased, the HR person hands it to the new employee to set up. Software engineers that use Linux have to find a USB drive and add their preferred distribution to the laptop. Anytime someone needs something from HR -- whether it's office related or tech related -- they email the HR representative directly.

When a new employee gets a machine, they're given logins to use cloud services. They get a personal orientation with HR to make sure they can login. This requires the HR person to block off a few hours for every new employee. If an employee forgets the login to their machine, they have no way to retrieve a password and they have to reimagine their machine. Employees don't have a strict password requirement to set for their computers.

The company currently has many of their services in the cloud, such as email, word processors, spreadsheet applications, etc. They also use the application, Slack, for instant communication.

Answer:

Here's a structured process review for Network Funtime Company based on their current IT infrastructure and restrictions:

Process Review – Network Funtime Company:

Network Funtime Company is a small open-source software firm with about 100 employees. Their IT infrastructure is minimal, with HR handling hardware purchases and onboarding. This has led to inefficiencies, security risks, and inconsistent setups. Below are five key improvements tailored to their size and financial constraints:

1. Standardized Hardware Procurement:

- Problem: HR buys random, cheapest laptops, leading to inconsistent performance and compatibility issues.
- Improvement: Create a standardized list of approved laptop models (e.g., one Linux-friendly developer model, one general-purpose model for non-technical staff).
- Rationale: Bulk purchasing or vendor partnerships reduces costs, ensures compatibility, and simplifies support.

2. Basic Asset Management System:

- Problem: No inventory tracking; stolen or missing devices cannot be audited.
- Improvement: Implement a lightweight asset management tool (e.g., open-source options like Snipe-IT) to log devices, assign them to employees, and track lifecycle.
- Rationale: Provides accountability, reduces losses, and supports audits without requiring expensive enterprise solutions.

3. Pre-Provisioned Devices:

- Problem: Employees wait days for laptops after hiring.
- Improvement: Maintain a small pool of pre-configured laptops ready for immediate deployment.

- Rationale: Reduces downtime, accelerates onboarding, and improves employee productivity from day one.

4. Centralized Onboarding & Password Management

- Problem: HR manually sets up accounts and spends hours per employee; forgotten passwords require full reimaging.

- Improvement: Use a centralized identity management system (e.g., Google Workspace or Azure AD) with self-service password reset and group-based provisioning.
- Rationale: Automates account creation, reduces HR workload, and improves security with password policies.

5. Security & Policy Enforcement:

- Problem: No password requirements, inconsistent OS setups, and ad-hoc installations.

- Improvement: Enforce minimum password complexity, enable disk encryption, and provide pre-configured OS images for Linux/Windows/Mac.

- Rationale: Improves security posture, ensures consistency, and reduces support overhead.

Conclusion:

By standardizing hardware, introducing lightweight asset management, pre-provisioning devices, centralizing onboarding, and enforcing security policies, Network Funtime Company can significantly improve efficiency and reduce risks. These solutions balance cost-effectiveness with scalability, ensuring the IT infrastructure supports growth without overwhelming resources.

Question 2:

Scenario 2:

You're doing systems administration work for W.D. Widgets. Evaluate their current IT infrastructure needs and limitations, then provide at least five process improvements and rationale behind those improvements. Please write a 200-400

word process review for this consultation. Remember, there's no right or wrong answer, but make sure to provide your reasoning.

Sales Company:

W.D. Widgets is a small company that sells widgets. They're mostly made up of salespeople who work with lots of clients. You've taken over as the sole IT person for this company of 80-100 people.

When HR tells you to provision a machine for a new employee, you order the hardware directly from a business vendor. You keep one or two machines in stock, in case of emergency. The users receive a username that you generate for them. You then give them an orientation on how to login when they start. You currently manage all of your machines using Windows Active Directory. The company uses only Windows computers. When a new computer is provisioned, you have to install lots of sales-specific applications manually onto every machine. This takes a few hours of your time for each machine. When someone has an IT-related request, they email you directly to help them.

Almost all software is kept in-house, meaning that you're responsible for the email server, local machine software, and instant messenger. None of the company's services are kept on the cloud.

Customer data is stored on a single file server. When a new salesperson starts, you also map this file server onto their local machine, so that they can access it like a directory. Whoever creates a folder on this server owns that folder and everything in it. There are no backups to this critical customer data. If a user deletes something, it may be lost for everyone.

The company generates a lot of revenue and is rapidly growing. They're expecting to hire hundreds of new employees in the next year or so, and you may not be able to scale your operations at the pace you're working.

Answer:

Process Review – W.D. Widgets:

W.D. Widgets is a growing sales company with 80–100 employees, relying heavily on in-house IT infrastructure. While the current setup works for a small team, it presents scalability and security challenges as the company prepares to expand rapidly. Below are five key improvements tailored to their needs and growth trajectory:

1. Automated Software Deployment:

- Problem: Each new machine requires manual installation of sales-specific applications, consuming hours of IT time.
- Improvement: Implement Windows Deployment Services (WDS) or Microsoft Endpoint Configuration Manager to automate OS imaging and application deployment.
- Rationale: Saves significant time, ensures consistency across devices, and scales easily as new hires increase.

2. Helpdesk Ticketing System:

- Problem: Employees email IT directly for support, creating inefficiency and lack of prioritization.
- Improvement: Deploy a lightweight ticketing system (e.g., Spiceworks or Freshservice).
- Rationale: Streamlines requests, tracks issues, and allows IT to prioritize and resolve problems systematically.

3. Data Backup and Recovery:

- Problem: Customer data resides on a single file server with no backups, risking catastrophic loss.
- Improvement: Implement automated backups to a secondary server or cloud-based storage, with versioning enabled.
- Rationale: Protects critical business data, ensures recovery from accidental deletions, and supports compliance.

4. Access Control and Permissions:

- Problem: Folder ownership is ad hoc, with no centralized control, leading to potential data silos and security risks.
- Improvement: Use Active Directory Group Policies to enforce role-based access and permissions.
- Rationale: Ensures data security, prevents unauthorized access, and supports collaboration without compromising integrity.

5. Cloud Migration Strategy:

- Problem: All services are in-house, which limits scalability and increases IT workload.
- Improvement: Gradually migrate non-critical services (e.g., email, instant messaging) to cloud providers like Microsoft 365 or Google Workspace.
- Rationale: Reduces infrastructure overhead, improves reliability, and allows IT to focus on core systems while scaling with growth.

Conclusion:

By automating deployments, introducing a helpdesk system, securing customer data with backups, enforcing access controls, and planning a phased cloud migration, W.D. Widgets can modernize its IT infrastructure. These improvements balance immediate efficiency with long-term scalability, ensuring the company can handle rapid expansion without overwhelming its sole IT administrator.

Question 3:

You're doing systems administration work for Dewgood. Evaluate their current IT infrastructure needs and limitations, then provide at least five process improvements and rationale behind those improvements. Please write a 200-400 word process review for this consultation. Remember, there's no right or wrong answer, but make sure to provide your reasoning.

Non-profit Company:

Dewgood is a small, local non-profit company of 50 employees. They hired you as the sole IT person in the company. The HR person tells you when they need a new computer for an employee. Currently, computers are purchased directly in a physical store on the day that an employee is hired. This is due to budget reasons, as they can't keep extra stock in the store.

The company has a single server with multiple services on it, a file server, and email. They don't currently have a messaging system in place. When a new employee is hired, you have to do an orientation with them for login. You're also responsible for installing all the software they need on their machine, and mapping the file server to their computer. The computers are managed through Windows Active Directory. When an employee leaves, they're currently not disabled in the directory service.

The company uses an open-source ticketing system to handle all internal requests as well as external non-profit requests. But the ticketing system is confusing and difficult to use, so lots of the employees reach out to you directly to figure out how to do things. In fact, so many things are difficult to find that employees typically ask around when they have a question.

There are nightly backups in place of the file server. You store this information on a disk backup and take it home with you everyday to keep it safe in case something happens onsite. There's also a small company website that's hosted on the single server at the company. This website is a single html page that explains the mission of the company and provides contact information. The website has gone down many times, and no one knows what to do when it happens.

Answer:

Process Review – Dewgood

Dewgood is a small non-profit with 50 employees, relying on a single server for multiple services and limited IT resources. While the current setup works at a basic level, it introduces risks and inefficiencies that could hinder operations. Below are five key improvements tailored to their budget-conscious environment:

1. Streamlined Hardware Procurement:

- Problem: Computers are purchased individually at local stores, causing delays and inconsistency.
- Improvement: Establish a relationship with a vendor to standardize hardware purchases and negotiate discounts.
- Rationale: Ensures compatibility, reduces costs, and avoids last-minute store runs while still respecting budget constraints.

2. Proper User Lifecycle Management: - Problem: Employees who leave are not disabled in Active Directory, creating security risks.

- Improvement: Implement a formal offboarding process where accounts are disabled immediately upon departure.
- Rationale: Protects sensitive data, prevents unauthorized access, and maintains directory hygiene.

3. Improved Ticketing System:

- Problem: The open-source ticketing system is confusing, leading employees to bypass it.
 - Improvement: Replace or reconfigure the system with a user-friendly option (e.g., OTRS, Freshdesk, or a simplified internal portal).
 - Rationale: Encourages proper use, reduces direct IT interruptions, and improves tracking of requests.
4. Resilient Backup Strategy:

- Problem: Backups are stored on physical disks taken home daily, which is risky and inefficient.
- Improvement: Introduce automated offsite or cloud-based backups with encryption.
- Rationale: Provides secure, reliable disaster recovery without relying on manual transport, while still being cost-effective.

5. Website Hosting Stability:

- Problem: The company's website is hosted on the same single server and frequently goes down.
- Improvement: Migrate the website to a low-cost external hosting provider.
- Rationale: Improves uptime, reduces server load, and ensures the public-facing mission statement remains accessible.

Conclusion By standardizing hardware procurement, enforcing user lifecycle management, improving ticketing usability, modernizing backups, and stabilizing website hosting, Dewgood can strengthen its IT infrastructure. These changes balance affordability with reliability, ensuring the non-profit can continue its mission without being hindered by preventable IT issues.